

INSuRE Spring 2023

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Privacy Within Federated Machine Learning Models

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Team 2

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1 Project Summary

1.1 Project Description

Federated Learning trains machine learning models on multiple devices, preserving data privacy and security by keeping data locally and only sharing model parameters. It enables training on a massive scale without compromising data privacy, and can occur on offline devices, making it suitable for resource-constrained environments like IoT devices or mobile phones. Federated Learning models communicate learnt parameters from local data sets to deliver better suggestions. However, privacy and data integrity are major concerns, and literature suggests possible defenses against these systems. FL trains models at different sites and updates weights from the training instead of transferring data to a central location, unlike classical machine learning.

1.2 Estimated Project Budget

This project is estimated to take around 500 hours of time and currently there are no monetary requirements.

1.3 Timeline of Milestones

Key Deliverables	Due Date
Project Proposal	2/6
Phase One	3/10
Midterm Presentation	3/10
Phase Two	3/11
Phase Three	4/7
Final Presentation and Report	5/5

1.4 Specific Aims

Our plan to tackle this problem involves analyzing the AAPFL model already in place, created by Minseok Ryu, and optimizing it further. While their model requires significantly less communication between the client and server and is communication efficient, we believe it can be improved by implementing a few major tweaks. Another approach is to develop an entirely new classifier that categorizes the given FL into either PII or PHI and provides a graphical model of the classification, which can describe the levels of privacy with reference to NIST SP800 series. In taking these approaches, we aim to achieve the following project-specific goals: 1) preserving privacy and not deanonymizing data, 2) determining which agent is attempting to retrieve data from the FL model, and 3) identifying the source of the privacy breach based on the information available and providing the privacy level using reference from NIST assurance levels.

2 Executive Summary

Federated learning is a transformative paradigm that brings many opportunities and challenges. Preserving privacy is critical and requires sophisticated techniques such as differential privacy, secure aggregation, and homomorphic encryption. However, these methods can be computationally expensive, so there is a need for ongoing research to develop more efficient privacy-preserving techniques for decentralized learning.

Despite challenges, investments in privacy-preserving techniques have led to promising developments, inspiring confidence in federated learning's future. By collaborating and exploring new solutions, we can create a future where federated learning drives innovation while safeguarding privacy and data security.

3 Previous Work

Privacy Preservation in Federated Learning: An insightful survey from the GDPR Perspective

With the rise of AI and machine learning-based applications and services, data privacy and security have become critical challenges. Typically, data is collected and aggregated in a data center where machine learning models are trained. However, this centralized approach has led to severe privacy risks, including personal data leakage, misuse, and abuse. Furthermore, in the era of the Internet of Things and big data, where data is essentially distributed, transferring vast amounts of data to a data center for processing is a cumbersome solution. This is not only due to difficulties in transferring and sharing data across data sources but also challenges in complying with rigorous data protection regulations and complicated administrative procedures, such as the EU General Data Protection Regulation (GDPR). Federated learning (FL) has emerged as a promising solution to this problem, facilitating distributed collaborative learning without disclosing original training data while naturally complying with the GDPR. Recent research has shown that retaining data and computation on-device in FL is insufficient for privacy guarantee. This article is dedicated to surveying the state-of-the-art privacy-preserving techniques that can be systematically employed in FL, as well as how these techniques mitigate data security and privacy risks.

Achieving security and privacy in federated learning systems: Survey, research challenges and future directions

Federated learning (FL) allows a server to learn a machine learning (ML) model across multiple decentralized clients that privately store their own training data. In contrast with centralized ML approaches, FL saves computation to the server and does not require the clients to outsource their private data to the server. However, FL is not free of issues. On the one hand, the model updates sent by the clients at each training epoch might leak information on the clients' private data. On the other hand, the model learnt by the server may be subjected to attacks by malicious clients; these security attacks might poison the model or prevent it from converging. In this paper, we first examine security and privacy attacks to FL and critically survey solutions proposed in the literature to mitigate each attack. Afterwards, we discuss the difficulty of simultaneously achieving security and privacy protection. Finally, we sketch ways to tackle this open problem and attain both security and privacy.

4 Motivation

While Federated Learning is designed to preserve data privacy, there are still some privacy concerns that need to be addressed. These include:

Model Inversion Attacks: In this type of attack, an adversary can reverse engineer the model parameters to gain access to the training data.

Model Stealing Attacks: Adversaries can steal the trained model and use it for malicious purposes.

Privacy Leaks through Model Parameters: The model parameters, which are shared during Federated Learning, can reveal information about the training data.

Unwanted data collection: Federated Learning requires data from multiple devices to be transmitted to a central server, which can lead to unwanted data collection and storage.

Model Biases: The model may learn biased representations due to the distribution of data on the devices.

Data Poisoning: An adversary can manipulate the training data on a device to poison the model parameters, which can lead to incorrect or malicious results.

To address these privacy issues, Federated Learning needs to be combined with privacy-preserving techniques such as differential privacy and secure multi-party computation to ensure that the training data is kept confidential and that the model parameters are kept secure.

Are there any more possible attacks on Federated Learning systems

Yes, there are several other possible attacks on Federated Learning systems, some of which include:

Model Poisoning: This type of attack involves introducing malicious updates to the federated model by an adversary, causing the model to behave abnormally and make incorrect predictions.

Model Stealing: In this type of attack, an adversary can extract the model's parameters after it has been trained on multiple parties' data, allowing them to access private information from the data.

Data Poisoning: An adversary can manipulate the data used to train the federated model by introducing false or misleading data, causing the model to make incorrect predictions.

Model Inversion: This type of attack involves using the federated model's predictions to infer sensitive information about the training data, potentially compromising the privacy of the participants.

Model Replacement: An adversary can replace the federated model with their own, potentially compromising the privacy of the participants or manipulating the model's predictions.

Backdoor Attacks: This type of attack involves inserting malicious code into the model, which allows an attacker to compromise the privacy of the participants or manipulate the model's predictions.

These attacks highlight the importance of robust privacy and security measures in Federated Learning systems, such as differential privacy and secure multi-party computation.

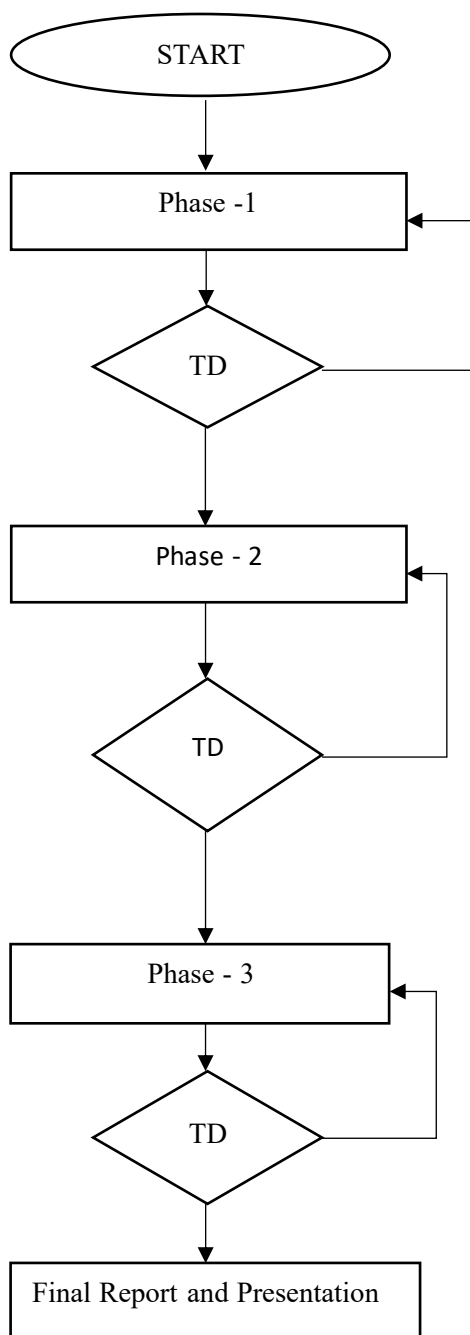
5 Research Project Plan and Approach

Phase 1: Get a basic understanding of Federated Learning, AAPFL model, and existing popular attacks on the machine learning models to see how federated learning models react.

Phase 2: Apply the data poisoning attack to the existing model and deploy counter measures. Analyze how to optimize the model and get the insights of privacy and data usage. Apply model inversion attack and try to reverse engineer the model parameters to get access to the data.

Phase 3: Refine the model and fine tune it to prevent such attacks and categorize the privacy of the model into levels.

Project Life Cycle:



6 Current Progress

We set up the APPFL framework and ran serial and MPI implementations of MNIST successfully. However, we encountered issues with GRPC and plan to test the framework with other datasets. Additionally, we plan to implement these samples on Tensorflow Federated and PySyft (PyTorch) too.

7 Deliverables

Key Deliverables	Due Date
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8 Anticipated Issues and Challenges

As most of us are beginners in the field of Machine Learning and its security applications, tackling big data sets and understanding the models would be a hurdle initially. We will try to learn different Federated ML models and how the data set parameters are used to increase the accuracy of the model.

The concept of Federated learning is relatively new. So, we need to get an in-depth knowledge of how they work, and we plan on performing few data poisoning attacks to see how the model reacts to that.

9 Investigator Biographies

Srinivas Panchangam holds a bachelor's degree in electronics and communications Engineering and is pursuing his masters at The University of Texas at Dallas, graduating in Dec 2023 with a focus in Data Science. He started his career as a software developer in one of the biggest tech companies in India, TCS. Though being from a different bachelors background his interest towards data science never ended, so he is trying to explore the field and get expertise in that.

With a bachelor's in Electronics and Communications Engineering and an Assistant System Engineer in TCS, Dhyana Sai Reddy Dalli is a second-year master's student in computer science specializing in Data Science. He has experience working in a variety of fields, such as health care, risk management, software development, Full stack Developer.

Neel Patel holds a bachelor's degree in information technology and is currently pursuing Master of Science in Computer Science at the University of Texas at Dallas with Information Assurance Track. He is NSA codebreaker 2022 solver. He possesses skills such as Binary Analysis, Software Reverse Engineering, Cryptography and programming skills in C++ and Python.

Lakshmi Saieesha Bonthala holds a bachelor's degree in Electronics and Communication Engineering and is currently pursuing her master's in computer science at The University of Texas at Dallas, with a focus in Data Science. She started her career in NTT Data as a Software Engineer with 2+ years of experience. She has done Machine Learning as a graduate course and is looking forward to gaining expertise in the field of Data Science. Gaining more exposure in cloud and machine learning areas is her main goal.

Nikhil Joshi is a MSCS Graduate Student at The University of Texas at Dallas. He is studying Cyber Security as my major. His coursework includes Network Security, Introduction to Cryptography, Malicious Code Analysis, Information Security, Data & Application Security, and Cloud Security. He regularly attends and practice CTF workshops and competitions. He actively looking for Full Time Roles in the field of Cyber Security and Computer Science.

Shrey Saini is a MSCS Graduate Student at The University of Texas at Dallas. He is studying Intelligence Systems as his major. He is well known for his knowledge in Python and Java Programming. He has immensely increased his knowledge space through his experience at Tata Consultancy Services where he worked as a systems Engineer. He actively looking for Full Time Roles in the field of Intelligence Systems and Computer Science.

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